

Estimation Theory

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Linear Models

Linear model: $\mathbf{x} = \mathbf{H}\theta + \mathbf{w}$, where $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$

uncorrelated noise
||

(white noise)

Example

$$x[n] = A + Bn + w[n] \quad n = 0, 1, \dots, N-1 \quad w[n] \sim \mathcal{N}(0, \sigma^2)$$

$$\begin{aligned} \mathbf{x} &= [x[0] \ x[1] \ \dots \ x[N-1]]^T \\ \mathbf{w} &= [w[0] \ w[1] \ \dots \ w[N-1]]^T \\ \boldsymbol{\theta} &= [A \ B]^T \end{aligned}$$

$$\mathbf{H} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ \vdots & \vdots \\ 1 & N-1 \end{bmatrix}$$

$$\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$$

Minimum Variance Unbiased

noiset gaussian
d2 gaussian

$$\hat{\theta} = \mathbf{g}(\mathbf{x}) \text{ is the MVU estimator if } \frac{\partial \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \mathbf{I}(\boldsymbol{\theta})(\mathbf{g}(\mathbf{x}) - \boldsymbol{\theta})$$

$$\frac{\partial \ln p(\mathbf{x}; \boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \frac{\partial}{\partial \boldsymbol{\theta}} \left[-\ln(2\pi\sigma^2)^{\frac{N}{2}} - \frac{1}{2\sigma^2} (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T (\mathbf{x} - \mathbf{H}\boldsymbol{\theta}) \right]$$

$$= -\frac{1}{2\sigma^2} \frac{\partial}{\partial \boldsymbol{\theta}} [\mathbf{x}^T \mathbf{x} - 2\mathbf{x}^T \mathbf{H}\boldsymbol{\theta} + \boldsymbol{\theta}^T \mathbf{H}^T \mathbf{H}\boldsymbol{\theta}]$$

$$= \frac{1}{\sigma^2} [\mathbf{H}^T \mathbf{x} - \mathbf{H}^T \mathbf{H}\boldsymbol{\theta}]$$

$$= \frac{\mathbf{H}^T \mathbf{H}}{\sigma^2} [(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{x} - \boldsymbol{\theta}] \quad (\text{if } \mathbf{H}^T \mathbf{H} \text{ is invertible})$$

$$\text{MVU estimator: } \hat{\boldsymbol{\theta}} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{x}$$

$$\hat{\boldsymbol{\theta}} \sim \mathcal{N}(\boldsymbol{\theta}, \sigma^2 (\mathbf{H}^T \mathbf{H})^{-1})$$

$$\mathbf{I}(\boldsymbol{\theta}) = \frac{\mathbf{H}^T \mathbf{H}}{\sigma^2}$$

$$\mathbf{C}_{\hat{\boldsymbol{\theta}}} = \mathbf{I}^{-1}(\boldsymbol{\theta}) = \sigma^2 (\mathbf{H}^T \mathbf{H})^{-1}$$

gaussian distribution

Example: Fourier Analysis for Periodic Signals

$$x[n] = \sum_{k=1}^M a_k \cos\left(\frac{2\pi kn}{N}\right) + \sum_{k=1}^M b_k \sin\left(\frac{2\pi kn}{N}\right) + w[n] \quad n = 0, 1, \dots, N-1$$

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$$

$$\boldsymbol{\theta} = [a_1 \ a_2 \ \dots \ a_M \ b_1 \ b_2 \ \dots \ b_M]^T$$

$$\mathbf{H} = \begin{bmatrix} 1 & \dots & 1 & 0 & \dots & 0 \\ \cos\left(\frac{2\pi}{N}\right) & \dots & \cos\left(\frac{2\pi M}{N}\right) & \sin\left(\frac{2\pi}{N}\right) & \dots & \sin\left(\frac{2\pi M}{N}\right) \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \cos\left[\frac{2\pi(N-1)}{N}\right] & \dots & \cos\left[\frac{2\pi M(N-1)}{N}\right] & \sin\left[\frac{2\pi(N-1)}{N}\right] & \dots & \sin\left[\frac{2\pi M(N-1)}{N}\right] \end{bmatrix}$$

$$\mathbf{H} = [\mathbf{h}_1 \ \mathbf{h}_2 \ \dots \ \mathbf{h}_{2M}]$$

Orthogonality (DFT):

$$\mathbf{h}_i^T \mathbf{h}_j = 0 \quad \text{for } i \neq j$$

$$\begin{aligned} \sum_{n=0}^{N-1} \cos\left(\frac{2\pi in}{N}\right) \cos\left(\frac{2\pi jn}{N}\right) &= \frac{N}{2} \delta_{ij} \\ \sum_{n=0}^{N-1} \sin\left(\frac{2\pi in}{N}\right) \sin\left(\frac{2\pi jn}{N}\right) &= \frac{N}{2} \delta_{ij} \\ \sum_{n=0}^{N-1} \cos\left(\frac{2\pi in}{N}\right) \sin\left(\frac{2\pi jn}{N}\right) &= 0 \quad \text{for all } i, j. \end{aligned}$$

if j is 0

Example: Fourier Analysis for Periodic Signals

$$x[n] = \sum_{k=1}^M a_k \cos\left(\frac{2\pi kn}{N}\right) + \sum_{k=1}^M b_k \sin\left(\frac{2\pi kn}{N}\right) + w[n] \quad n = 0, 1, \dots, N-1$$

$$\begin{aligned} \mathbf{H}^T \mathbf{H} &= \begin{bmatrix} \mathbf{h}_1^T \\ \vdots \\ \mathbf{h}_{2M}^T \end{bmatrix} [\mathbf{h}_1 \ \dots \ \mathbf{h}_{2M}] \quad \text{orthogonality} \quad \frac{2}{N} \quad 0/\frac{N}{2} \\ &= \begin{bmatrix} \mathbf{h}_1^T \mathbf{h}_1 & \mathbf{h}_1^T \mathbf{h}_2 & \dots & \mathbf{h}_1^T \mathbf{h}_{2M} \\ \mathbf{h}_2^T \mathbf{h}_1 & \mathbf{h}_2^T \mathbf{h}_2 & \dots & \mathbf{h}_2^T \mathbf{h}_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{h}_{2M}^T \mathbf{h}_1 & \mathbf{h}_{2M}^T \mathbf{h}_2 & \dots & \mathbf{h}_{2M}^T \mathbf{h}_{2M} \end{bmatrix} = \begin{bmatrix} \frac{N}{2} & 0 & \dots & 0 \\ 0 & \frac{N}{2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{N}{2} \end{bmatrix} = \frac{N}{2} \mathbf{I} \end{aligned} \quad \mathbf{h}_i^T \mathbf{h}_j = 0 \quad \text{for } i \neq j$$

MVU estimator: $\hat{\boldsymbol{\theta}} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{x}$

$$\begin{aligned} &= \frac{2}{N} \mathbf{H}^T \mathbf{x} = \frac{2}{N} \begin{bmatrix} \mathbf{h}_1^T \\ \vdots \\ \mathbf{h}_{2M}^T \end{bmatrix} \mathbf{x} \\ &= \begin{bmatrix} \frac{2}{N} \mathbf{h}_1^T \mathbf{x} \\ \vdots \\ \frac{2}{N} \mathbf{h}_{2M}^T \mathbf{x} \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \hat{a}_k &= \frac{2}{N} \sum_{n=0}^{N-1} x[n] \cos\left(\frac{2\pi kn}{N}\right) \\ \hat{b}_k &= \frac{2}{N} \sum_{n=0}^{N-1} x[n] \sin\left(\frac{2\pi kn}{N}\right) \end{aligned}$$

Linear model: $\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$, where $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{C})$

(colored noise)

$$\mathbf{C}^{-1} = \mathbf{D}^T \mathbf{D} \quad ; \text{covariance matrix is positive definite}$$

D: whitening transformation

$$E[(\mathbf{D}\mathbf{w})(\mathbf{D}\mathbf{w})^T] = \mathbf{D}\mathbf{C}\mathbf{D}^T$$

$$= \cancel{\mathbf{D}}\cancel{\mathbf{D}^{-1}}\cancel{\mathbf{D}^{T^{-1}}}\cancel{\mathbf{D}^T} = \mathbf{I} \Rightarrow$$

$\mathbf{D}\mathbf{w}$ is white noise

$$\begin{aligned} \mathbf{x}' &= \mathbf{D}\mathbf{x} \\ &= \mathbf{D}\mathbf{H}\boldsymbol{\theta} + \mathbf{D}\mathbf{w} \\ &= \mathbf{H}'\boldsymbol{\theta} + \mathbf{w}' \quad \mathbf{w}' = \mathbf{D}\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \end{aligned}$$

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= (\mathbf{H}'^T \mathbf{H}')^{-1} \mathbf{H}'^T \mathbf{x}' \\ &= (\mathbf{H}^T \mathbf{D}^T \mathbf{D} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{D}^T \mathbf{D} \mathbf{x} \end{aligned}$$

$$\mathbf{C}_{\hat{\boldsymbol{\theta}}} = (\mathbf{H}'^T \mathbf{H}')^{-1}$$

MVU estimator:

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T \mathbf{C}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{C}^{-1} \mathbf{x}$$

$$\mathbf{C}_{\hat{\boldsymbol{\theta}}} = (\mathbf{H}^T \mathbf{C}^{-1} \mathbf{H})^{-1}$$

General Linear Model:

$$x(t) = \sum_{i=1}^M \theta_i h_i(t) + w(t), \quad w(t) \sim \mathcal{N}(0, \sigma^2)$$

$h_i[t]$: nonlinear function of t

$$h_i(t) = t^{i-1}$$

$$h_i(t) = \exp\left(-\frac{1}{2s^2}(t - m_i)^2\right)$$

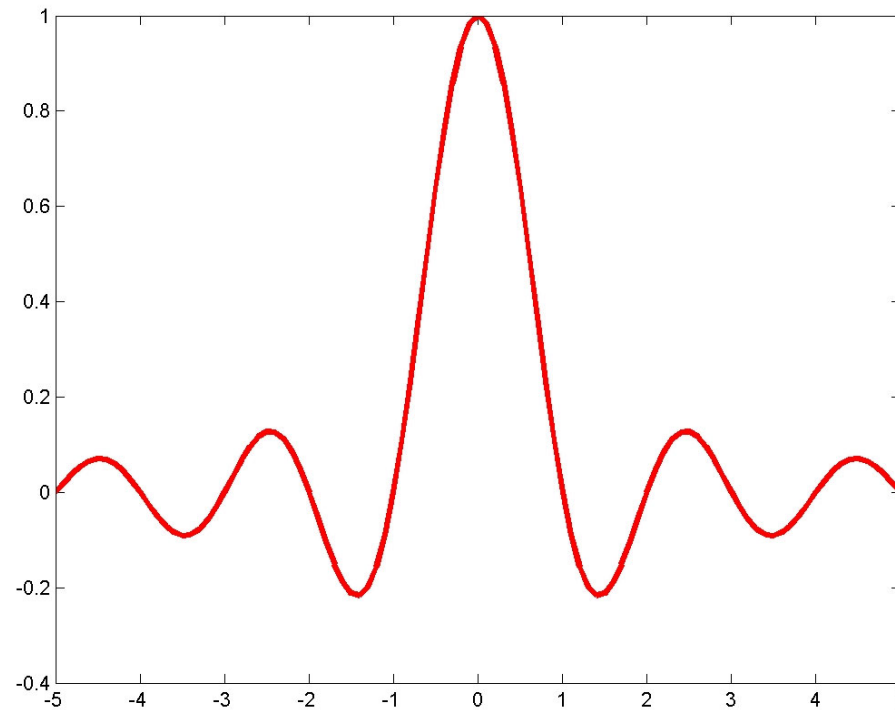
$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$$

$$\mathbf{H} = \begin{bmatrix} h_1(t_0) & h_2(t_0) & \cdots & h_M(t_0) \\ h_1(t_1) & h_2(t_1) & \cdots & h_M(t_1) \\ \vdots & \vdots & \ddots & \vdots \\ h_1(t_{N-1}) & h_2(t_{N-1}) & \cdots & h_M(t_{N-1}) \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_M \end{bmatrix} \quad \mathbf{w} = \begin{bmatrix} w(t_0) \\ w(t_1) \\ \vdots \\ w(t_{N-1}) \end{bmatrix}$$

$$\text{MVU Estimator: } \hat{\boldsymbol{\theta}} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{x}$$

Example: Linear modeling of the SINC function



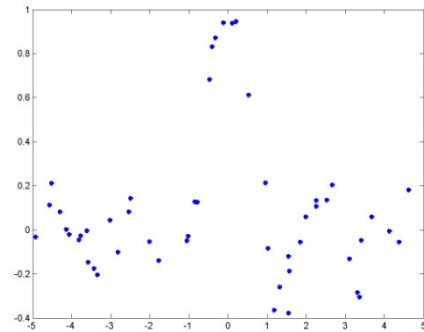
Model 1: $x(t) = A + Bt + w(t)$, $w(t) \sim \mathcal{N}(0, \sigma^2)$

Model 2: $x(t) = \sum_{i=1}^M \theta_i h_i(t) + w(t)$, $w(t) \sim \mathcal{N}(0, \sigma^2)$ $h_i(t) = \exp\left(-\frac{1}{2s^2}(t - m_i)^2\right)$

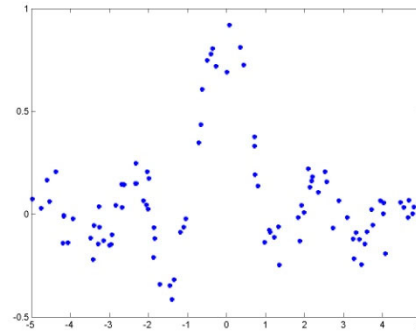
Example: Linear modeling of the SINC function

Data

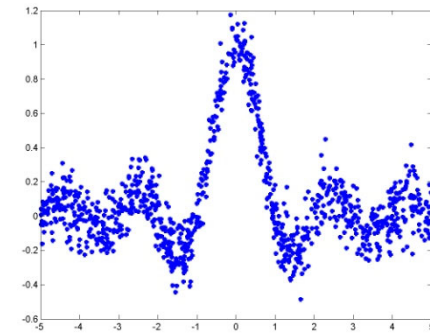
N=50



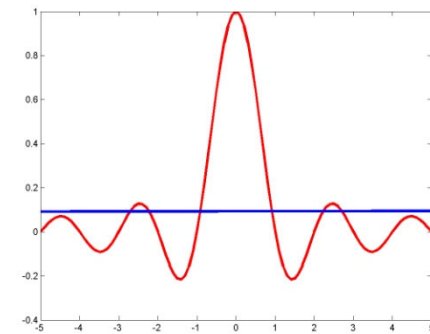
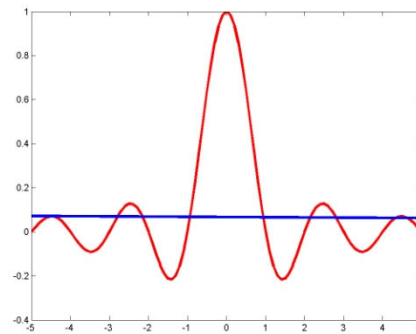
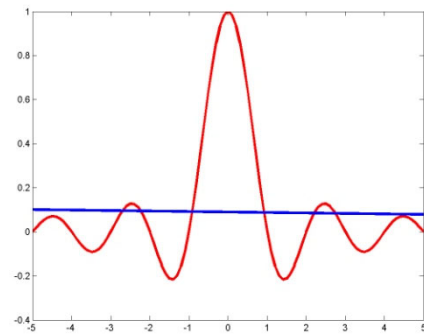
N=100



N=1000



Linear
Model 1



Linear
Model 2

